

Product Requirements Document

Product: 3DUp

Version: MVP v1

1. Product Overview

3DUp is a **project and asset management platform for real estate 3D production**.

It allows real estate teams to:

- Organize project assets (blueprints, requirements, moodboards)
- Share them with freelancers in a structured workspace
- Receive uploaded 3D renders organized by apartment, room, and design tier

The goal is to **eliminate fragmented workflows across email, WhatsApp, Google Drive, and WeTransfer** and replace them with a **single structured workspace per project**.

2. Problem Statement

Producing 3D renders for real estate projects typically involves:

- Sending blueprints via email
- Sharing references in Google Drive
- Explaining requirements in WhatsApp
- Receiving renders via WeTransfer

This results in:

- lost files
- unclear asset structure
- difficult revisions
- poor organization across multiple projects

3DUp solves this by providing **one structured platform where assets and outputs are organized and accessible.**

3. Goals of the MVP

The MVP must allow teams to:

1. Create projects
2. Upload project assets
3. Structure apartment types
4. Provide freelancers with easy access to assets
5. Allow freelancers to upload renders
6. Allow admins to review outputs

The system should prioritize:

- simplicity
 - clarity
 - file organization
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4. Users

Admin (internal team)

Responsibilities:

- Create projects
 - Upload building assets
 - Upload apartment blueprints
 - Upload design moodboards
 - Invite freelancers
 - Review uploaded renders
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Freelancer (3D artist)

Responsibilities:

- Review project assets
- Download blueprints and moodboards
- Upload rendered panoramas

Freelancers cannot edit project assets.

5. Core Entities

The platform is structured around four main objects.

Project

Represents a real estate development.

Contains:

- building assets
 - apartments
 - freelancers
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Apartment Type

Represents a specific apartment model within the project.

Contains:

- blueprint
 - moodboards
 - rooms
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Room

Represents a space within an apartment.

Examples:

- Living room
- Kitchen

- Bedroom
- Bathroom

Rooms contain render outputs.

Render Tier

Each room contains **three render tiers**.

Tier 1

Tier 2

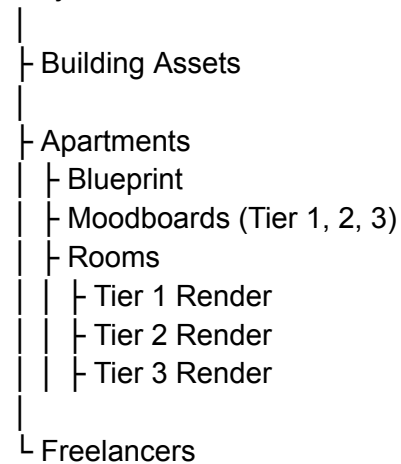
Tier 3

These represent different interior design levels.

6. Information Architecture

Projects

Project



7. MVP Features

The MVP includes the following modules:

1. Authentication
 2. Project management
 3. Asset uploads
 4. Freelancer workspace
 5. Render upload system
 6. Review system
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8. User Flows

Flow 1

Admin Creates Project

Admin navigates to:

Projects → Create New Project

Step 1

Enter project details

Fields:

- Project name
- Project description

System creates project container.

Flow 2

Upload Building Assets

Admin uploads assets related to the building.

Fields:

- Building logo
- Building images
- Drone video (optional)

Assets are stored under:

Project → Building Assets

Flow 3

Add Apartment Types

Admin creates apartment types within the project.

For each apartment:

Fields:

- Apartment name
- Apartment description
- Blueprint upload

Admin can add multiple apartment types.

Example:

Apartment A
Apartment B
Apartment C

Flow 4

Upload Apartment Moodboards

Each apartment requires **three moodboards**.

Tier 1 moodboard
Tier 2 moodboard
Tier 3 moodboard

These moodboards guide the 3D designer.

Location:

Project → Apartment → Moodboards

Flow 5

Define Rooms

Admin defines the rooms that need rendering.

Example rooms:

- Living room
- Kitchen
- Bedroom
- Bathroom

Rooms determine the freelancer upload structure.

Flow 6

Assign Freelancer

Admin invites freelancer to project.

Steps:

Admin clicks "Invite Freelancer".

Fields:

- Freelancer name
- Email

Freelancer receives access to the project workspace.

Flow 7

Freelancer Reviews Assets

Freelancer logs in.

Freelancer dashboard shows assigned projects.

Freelancer opens project.

Freelancer sees:

Building Assets

- building images
- logo
- references

Apartment Assets

- blueprint
- moodboards

All files are downloadable.

Flow 8

Freelancer Uploads Renders

Freelancer selects:

Project → Apartment → Room

Upload slots exist for:

Tier 1 render

Tier 2 render

Tier 3 render

Each slot accepts:

- panorama images
- high resolution render images

Uploaded files appear immediately in the admin workspace.

Flow 9

Admin Reviews Uploads

Admin opens project.

Navigates to:

Apartment → Room

Admin sees uploaded renders.

Admin can:

- preview renders
 - download renders
 - request revisions (optional future feature)
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9. Screens Required

The MVP should include the following screens.

1. Login
2. Projects dashboard
3. Create project flow

4. Project overview
 5. Apartment detail page
 6. Freelancer workspace
 7. Render upload interface
 8. Freelancer management page
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10. Upload Requirements

Supported file types:

Images:

jpg
png
webp

Documents:

pdf

Video:

mp4

Render files may be large, so upload component must support:

- drag and drop
 - preview
 - replace file
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11. Permissions

Admin:

- create projects
- upload assets
- invite freelancers
- review renders

Freelancer:

- view assets
- download assets
- upload renders

Freelancers cannot edit assets.

12. Non Functional Requirements

The system should:

- handle large file uploads
- provide fast download access
- support multiple projects simultaneously
- store files securely

Recommended stack:

Frontend: React / Next.js

Backend: Supabase

Storage: Supabase Storage or S3

13. Success Metrics

The MVP is successful if users can:

- upload project assets in under 5 minutes
 - freelancers can find required files immediately
 - renders are uploaded in the correct structure
 - projects remain organized
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14. Future Features (Post MVP)

These are **not part of the MVP** but should be considered later.

- render approval workflow
 - comments on renders
 - revision tracking
 - version history
 - progress status tracking
 - integration with 3D configurator platforms
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Final Summary

3DUp MVP should deliver a **simple structured workspace where real estate teams organize project assets and freelancers upload renders in a predictable structure.**

The core principle is:

Clear structure → Easy access → Organized outputs

If you'd like, I can also create the **database schema (Supabase tables)** and **API structure**, which will make Cursor or Claude build this platform **much more reliably.**